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Project Title:	HZE Radiation: Modulation Of Genetic Effects By RNA Interference Of NHEJ		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Bedford, Joel (Colorado State University) Liber, Howard (Colorado State University)		
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Task Description:

For extended missions in space, astronauts will be exposed to a radiation environment unlike any on the earth's surface. Galactic cosmic rays include a particularly worrisome component of energetic iron nuclei which are high atomic number, high energy (HZE) charged particles with linear energy transfer (LET) similar to alpha particles but with very different track structures. While the relative biological effectiveness (RBE) of a particular radiation roughly depends on the LET, the very different track structures for these two radiations may well modulate this RBE even if they have identical average LETs. Further, this "radiation quality" for the exposures actually relevant for astronauts will be modified by interactions with materials in the spacecraft, including their own bodies.
Basic to virtually all relevant biological effects of ionizing radiation is the underlying damage produced in DNA, and the subsequent cellular processing of such damage. The damage can be qualitatively different for different kinds of radiations, and the genetics of the biological systems exposed can greatly affect damage processing and ultimate outcome—the biological effect of concern. The accurate repair of DNA double-strand breaks (DSB) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosome rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. It is important, therefore, to understand cellular responses to DNA damage and to elucidate the mechanisms and phenotypic consequences of DSB misrepair.

Our general aim in this proposal is to examine connections between cellular radiosensitivity phenotypes relevant for carcinogenesis, namely chromosome aberration formation, mutagenesis and telomere fusions, and deficiencies in various components of the non-homologous end joining system, using RNA interference to systematically knockdown gene products known to be involved, either singly or in combination.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

This project utilizes the relatively new technique of RNA interference (RNAi) to deplete levels of various components of the non-homologous end joining (NHEJ) double-strand break (DSB) repair system, and to study the effects on HZE radiation-induced chromosomal rearrangements and gene mutation.
The accurate repair of DNA double-strand breaks (DSBs) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosomal rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. We have focused on the consequent phenotypic effects of faulty repair by examining connections between cellular radiosensitivity phenotypes relevant for carcinogenesis after exposure to IR, and deficiencies in various components of the non-homologous end joining (NHEJ) system. We produced deficiencies of individual components of the DNA-dependent protein kinase (DNA-PK) holoenzyme (Ku86 and the catalytic subunit, DNA-PKcs), both singly and in combination, using RNA interference (RNAi) in human lymphoblastoid cell lines. Exposure of cells exhibiting reduced protein expression to either gamma-rays or 1GeV/nu Fe particles demonstrated differential effects on telomere dysfunction and mutation frequency, as well as differential effects between radiation qualities. Our initial results are reported in the article "Suppression of DNA-PK by RNAi has different quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma-rays or HZE particles" by Qinming Zhang, Eli S. Williams, Kristin F. Askin, Yuanlin Peng, Joel S. Bedford, Howard L. Liber and Susan M. Bailey, Radiation Research, 164:497-504 (NASA supplement).

Mutations of the NBS1 gene are responsible for Nijmegen breakage syndrome (NBS), a rare autosomal recessive hereditary disorder that imparts an increased predisposition to development of malignancy. The NBS1 protein is a component of the MRE11/RAD50/NBS1 (MRN) complex that in turn plays a critical role in the regulation of cellular responses to DNA damage and thus the maintenance of chromosomal integrity. NBS has been proposed to function in both NHEJ and HR. Due to the fact that NBS1 is not a component of NHEJ, but still critical for DNA repair, we decided to look at the effects of a deficiency of NBS1 on the consequences of IR. Use of our now standard siRNA technique resulted in a 95% reduction in NBS1 protein 5 days after transfection into WTK1 cells. The cells showed impaired phosphorylation of H2AX, as well as impaired activation of the cell cycle checkpoint regulating kinase, Chk2. These results thus provided evidence that the knockdown was effective on a functional level. We observed an increased level of gamma-ray-induced mutation at two genetic loci. NBS1 deficiency also resulted in a significant increase in telomere association that was independent of radiation exposure. Our results provide the first experimental evidence that NBS1 deficiency in human cells leads to hypermutability and telomere associations, phenotypes that may contribute to the cancer predisposition seen among patients with this disease. This work has recently been published: Zhang, Y., C.-U.K. Lim, E.S. Williams, J. Zhou, Q. Zhang, M.H. Fox, S.M. Bailey and H.L. Liber (2005). NBS1 knockdown by siRNA increases ionizing radiation mutagenesis and telomere association in human cells. Cancer Research, 65:5545-5554.

Future experiments will be done with HZE particles. Two experiments were done at Brookhaven National Laboratory in the past year, in April and June 2005. These experiments were designed to further explore the dose-response kinetics of HZE particle-induced chromosome changes and mutagenesis.

We conducted one experiment using mFISH to monitor chromosome aberration formation in WTK1 cells treated with 1 Gy of 56Fe ions. This experiment was done with or without knockdown of DNA-PKcs. We made the following preliminary observations: 1) The total number of break points was the same (with or without knockdown); HZE exposure produced the same number of break points, regardless of DNA-PKcs background. 2) There was an increased number of "open breaks" (terminal deletions, incomplete exchanges) with the knockdown; we interpret this as reflecting the repair deficiency associated with reduced DNA-PKcs levels, especially in dealing with HZE-induced DSBs. 3) There was a decreased number of complex exchanges and increased number of simple exchanges with knockdown; this may be indicative of the alternate/backup repair system being utilized in the absence of sufficient DNA-PKcs.

These initial and intriguing results need to be repeated. These studies also need to be done in conjunction with telomere FISH, the logistics of which we are working on. They also need to be done with gamma-rays to compare and contrast.

Our experiments will lead to valuable information regarding CPR question 10.12, “What are the effects of inter-individual diversity?” We hypothesize that NHEJ repair components are major genetic determinants controlling individual susceptibility to radiation-induced cancer, as well as heritable effects of exposure. Therefore, naturally occurring deficiencies of these proteins within the population will put these individuals at increased risk. We also anticipate addressing CPR question 10.07, “What are the cancer risks in humans from spaceflight?” We anticipate that our characterization of telomere dysfunction, chromosome aberration formation and gene mutation will serve as surrogate endpoints for predicting cancer risks.

Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Zhang Y, Lim CU, Williams ES, Zhou J, Zhang Q, Fox MH, Bailey SM, Liber HL. "NBS1 knockdown by small interfering RNA increases ionizing radiation mutagenesis and telomere association in human cells." Cancer Res. 2005 Jul 1;65(13):5544-53. PMID: 15994926 , Jul-2005
Articles in Peer-reviewed Journals	Zhang Q, Williams ES, Askin KF, Peng Y, Bedford JS, Liber HL, Bailey SM. "Suppression of DNA-PK by RNAi has different quantitative effects on telomere dysfunction and mutagenesis in human lymphoblasts treated with gamma rays or HZE particles." Radiat Res. 2005 Oct;164(4 Pt 2):497-504. PMID: 16187756 , Oct-2005